Brandon Nguyen - Pathway 1: Sensing the World Reflection Due Date: 09/9/2026

You are expected to type your reflection in essay format. The reflection should start with a personalized summary of what was done that week and how it was accomplished. Then you will describe what you learned and your experience. This should all be from your perspective, so pronouns are expected.

During the first week of labs, we were introduced to several foundational engineering concepts and assigned specific hands-on tasks to complete during the practical portion of the module. Our primary goal was to determine and compare the electrical conductivity of silver (Ag) paste, silver paint, and carbon paint. Within my three-person group, I recorded the data we collected and managed the hot plate once the paint had dried, while the other two focused on applying the paints and peeling the delicate electrode layer needed to transfer the paint.

We began the experiment by peeling the electrode layer to adhere it onto a glass slide. Although this preparation took many attempts due to the fragility of the material, my partner successfully prepared three distinct slides. My other partner then applied dabs of the Ag paint onto the glass slide and squeegeed it into a thin layer. Upon finishing the thin layer, I'd carefully place it on top of the hot plate as we waited a few minutes for it to dry. Upon completion, we'd let it cool down for a minute before testing its resistance. As my partner tested the resistance for the Ag paint, I began jotting down the data we gathered, which was .025 Ω. We would then use the measurement of the thickness of the tape we found using a caliper and divide it to find the resistivity, which was .000625 Ωmm. However, since we were asked to find the conductivity of the material, we used the formula $\sigma = 1/p$ which we found was 1600 1/(Ωm). We would then redo the experiment with our two remaining materials, as we found the conductivity of Ag paste to be 2.667 1/(Ωm), and carbon paint, .533 1/(Ωm).

In the next lab, we would then test its conductivity after putting it through a spin-coat of PEDOT-PSS. We used our Ag paint sample as our main sample for this experiment, as one of my partners went to get our glass slide spin-coated with PEDOT. After coating it, we tested the resistivity of the PEDOT-PSS film in ambient air. Measuring its resistivity as 17.5 Ωmm, we then began to breathe on the film as we were tasked to do, retested it, and recalculated it. After finishing the calculations, we found that the resistivity of the PEDOT film spiked sharply to 91 Ωmm, representing an approximate 420% increase in resistance due to humidity.

From what I've learned throughout these experiments, there will be many roadblocks, and the labs will rarely go perfectly according to plan. This was especially true for my group, as the first lab was very overwhelming and sorta confusing. But as we continued to go to the labs and get assistance from the TAs, we were able to increase our productivity and efficiency. This experience taught me, as an aspiring engineer, to always put my ego aside and be willing to ask for help when needed, because being an engineer isn't a solo job but a job that needs effective communication between co-workers and a willingness to continue learning, as after all, engineers are only as strong as their unit. And I think the time we were given, while it may be detrimental to our work, created an effective line of communication and a willingness to cooperate between the three of us, as none of us were willing to fall behind on these assignments and all of us were ready to communicate, share our opinions, and help each other out if it meant fulfilling our quotas for the lab.

I also learned how limited lab time truly is. My team and I often didn't finish some parts of the lab just because another group returned early, which meant we had less time to work. As a result, the beginning of each lab was usually spent catching up so we would not fall behind. And this finite amount of time in the lab told me just how critical time management is in engineering

and how important it is to use limited time efficiently. Overall, I learned several ways to calculate resistivity and conductivity using tools such as a multimeter, calipers, and a variety of materials.

In addition to the labs, I greatly enjoyed the guided tours of our campus engineering facilities, particularly the cleanrooms and design labs. Exploring spaces ranging from the 3D printing hub to the metal workshops opened my eyes to resources that I, as a freshman, can leverage throughout my time here at Lehigh. Overall, I'd say the hands-on approach for this module has been a great benefit and has definitely got me interested in the electrical engineering and coding aspects of today's society. And while I may be leaning towards bioengineering, the module so far has definitely shown me ways in which advanced chips and electrical engineering could greatly benefit today's healthcare and our equipment for the better.